# Breast Cancer Analysis

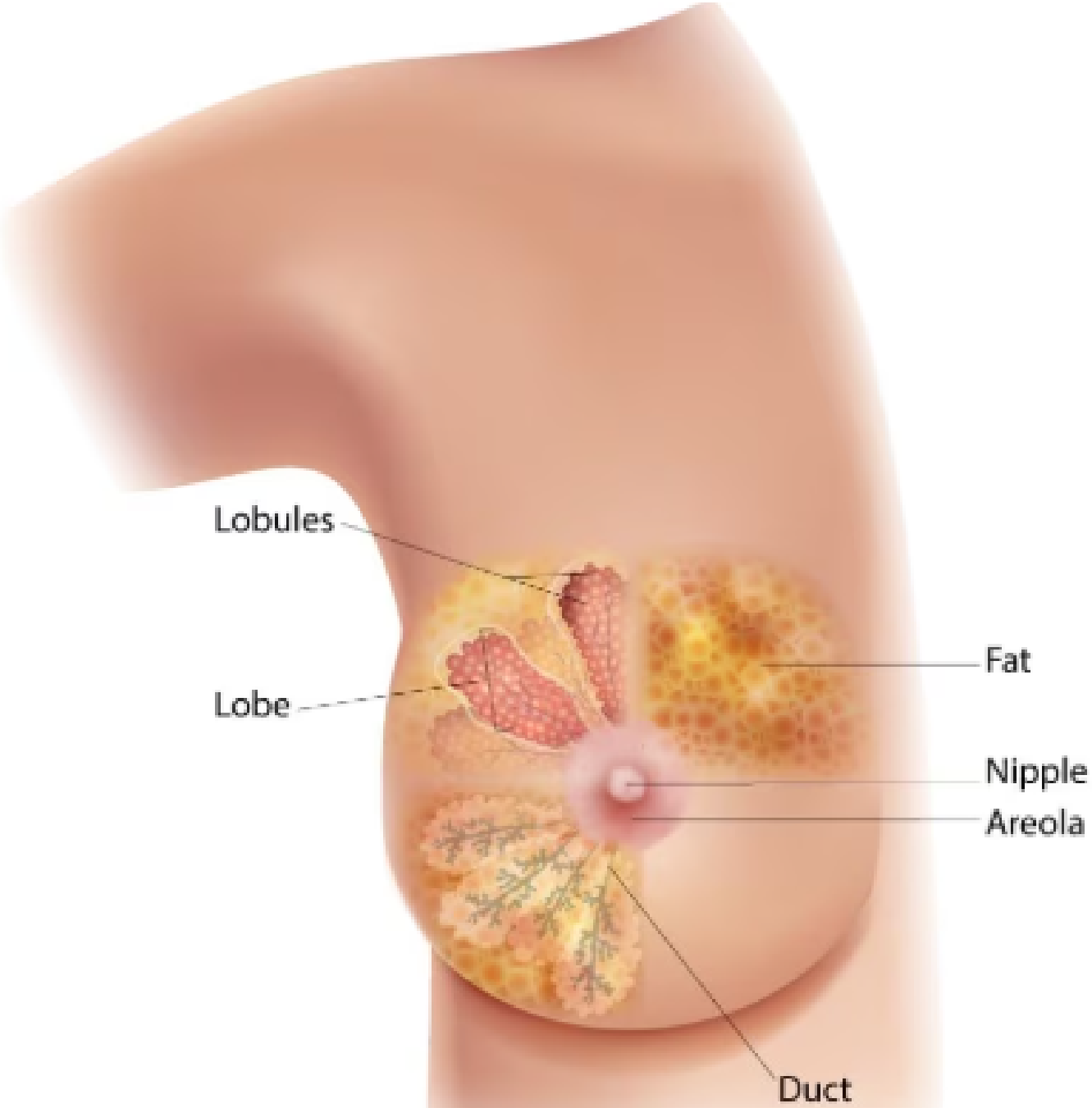
Name

Submitted On: Feb 5, 2025

# Introduction

- **Significance of Breast Cancer as a Global Health Issue**
  - In 2022, approximately **2.3 million women** were diagnosed with breast cancer worldwide.
  - This represents about **one in four** of all cancers in women.
  - In 2022, **670,000 women** died from breast cancer worldwide.
  - In 2020, breast cancer was the cause of **16%** of cancer deaths in women.
  - Breast cancer incidence has increased by more than **20%** since 2008.
  - Breast cancer incidence is growing in many parts of the world, especially in transitioning countries.
  - In 2019, **25,100 men** were diagnosed with breast cancer globally.

# Breast Cancer Overview Image



# Brief intro to types of breast cancer

**Benign vs. Malignant Tumors**

| Category | Description |
|---|---|
| **Benign Tumors** | Non-cancerous, usually slow-growing, doesn't spread, usually not life-threatening. |
| **Malignant Tumors** | Cancerous, can grow rapidly, invade surrounding tissues, spread to other body parts, potentially life-threatening. |

# Brief intro to types of breast cancer

**Invasive Breast Cancers: Types**

| **Feature** | **Invasive Ductal Carcinoma (IDC)** | **Invasive Lobular Carcinoma (ILC)** |
|---|---|---|
| **Frequency** | Most common type of breast cancer (70-80%). | Second most common (10-15%). |
| **Origin** | Starts in the milk ducts. | Starts in the milk lobules. |
| **Presentation** | May form a lump and may spread to other tissues. | May not typically form a lump; presents as a thickening and may affect both breasts. |

# Project Motivation

- Chose this topic because breast cancer is a major issue.
- Goals of the project:
  - Analyze key datasets related to breast cancer.
  - Apply statistical methods to uncover significant insights.
  - Inform strategies for early detection and treatment.

# Dataset Introduction

**Source of the Dataset:**

This breast cancer domain was obtained from the University Medical Centre, Institute of Oncology, Ljubljana, Yugoslavia. This is one of three domains provided by the Oncology Institute that has repeatedly appeared in the machine learning literature.

**Citations:**

Zwitter, M. & Soklic, M. (1988). Breast Cancer [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C51P4M

**Download Link:**

https://archive.ics.uci.edu/dataset/14/breast+cancer

# Breast Cancer Diagnosis Data Preview

| id | diagnosis | radius_mean | texture_mean |
|---|---|---|---|
| 842302 | M | 17.990000 | 10.380000 |
| 842517 | M | 20.570000 | 17.770000 |
| 84300903 | M | 19.690000 | 21.250000 |

| perimeter_mean | area_mean | smoothness_mean | compactness_mean |
|---|---|---|---|
| 122.800000 | 1001.000000 | 0.118400 | 0.277600 |
| 132.900000 | 1326.000000 | 0.084740 | 0.078640 |
| 130.000000 | 1203.000000 | 0.109600 | 0.159900 |

# Breast Cancer Diagnosis Data Preview (Part 3)

| concavity_mean | concave points_mean | symmetry_mean |
|---|---|---|
| 0.300100 | 0.147100 | 0.241900 |
| 0.086900 | 0.070170 | 0.181200 |
| 0.197400 | 0.127900 | 0.206900 |

| fractal_dimension_mean | radius_se | texture_se |
|---|---|---|
| 0.078710 | 1.095000 | 0.905300 |
| 0.056670 | 0.543500 | 0.733900 |
| 0.059990 | 0.745600 | 0.786900 |

# Breast Cancer Diagnosis Data Preview (Part 4)

| perimeter_se | area_se | smoothness_se |
|---|---|---|
| 8.589000 | 153.400000 | 0.006399 |
| 3.398000 | 74.080000 | 0.005225 |
| 4.585000 | 94.030000 | 0.006150 |

| compactness_se | concavity_se | concave points_se |
|---|---|---|
| 0.049040 | 0.053730 | 0.015870 |
| 0.013080 | 0.018600 | 0.013400 |
| 0.040060 | 0.038320 | 0.020580 |

# Breast Cancer Diagnosis Data Preview (Part 5)

| symmetry_se | fractal_dimension_se | radius_worst |
|---|---|---|
| 0.030030 | 0.006193 | 25.380000 |
| 0.013890 | 0.003532 | 24.990000 |
| 0.022500 | 0.004571 | 23.570000 |

| texture_worst | perimeter_worst |
|---|---|
| 17.330000 | 184.600000 |
| 23.410000 | 158.800000 |
| 25.530000 | 152.500000 |

# Global Cancer Data Preview

| Label | Cancer code | Country code (ISO/UN) | Alpha-3 code | Sex |
|---|---|---|---|---|
| Lip oral cavity | 1 | 900 | N.A. | 0 |
| Salivary glands | 2 | 900 | N.A. | 0 |
| Oropharynx | 3 | 900 | N.A. | 0 |

| Type | ASR (World) | Crude rate | Cumulative risk | Total |
|---|---|---|---|---|
| 0 | 4.000000 | 4.940000 | NaN | 389846 |
| 0 | 0.560000 | 0.700000 | NaN | 55083 |
| 0 | 1.100000 | 1.350000 | NaN | 106400 |

# Breast Cancer Statistics by Country Preview

| Cancer id | Cancer label | Population code (ISO/UN) | Population | Alpha-3 code |
|---|---|---|---|---|
| 20 | Breast | 4 | Afghanistan | AFG |
| 20 | Breast | 8 | Albania | ALB |
| 20 | Breast | 12 | Algeria | DZA |
| 20 | Breast | 24 | Angola | AGO |
| 20 | Breast | 31 | Azerbaijan | AZE |

| Sex | Type | ASR (World) per 100 000 |
|---|---|---|
| 0 | 0 | 29.440000 |
| 0 | 0 | 51.140000 |
| 0 | 0 | 61.870000 |
| 0 | 0 | 29.430000 |
| 0 | 0 | 32.900000 |

# Breast Cancer Continent Mortality Data Preview

| Continents | Cancer code | Country code (ISO/UN) | Alpha-3 code | Sex |
|---|---|---|---|---|
| Africa | 20 | 903 | N.A. | 0 |
| Latin America and the Caribbean | 20 | 904 | N.A. | 0 |
| Northern America | 20 | 905 | N.A. | 0 |

| ASR (World) | Crude rate | Cumulative risk | Total |
|---|---|---|---|
| 19.160000 | 12.970000 | NaN | 91252 |
| 13.210000 | 17.700000 | NaN | 59876 |
| 12.320000 | 26.380000 | NaN | 49744 |

# Statistical Measures and Tools - Introduction

**Statistical Tools for Data Analysis**

In this section, we will introduce the statistical measures, concepts, and visualization tools employed to analyze the breast cancer datasets. These tools help us to:

- Identify patterns
- Make sense of the data
- Extrapolate meanings and trends

# Statistical Measures and Tools (Part 1)

**Categories of Statistical Tools Used: Descriptive Statistics (Part 1)**

- Mean
- Median
- Mode
- Minimum/S
- Maximum/L
- Range
- Coefficient of Range
- MD Mean
- MD Median
- Standard Deviation
- Variance
- Coefficient of SD
- CV
- IQR
- QD
- Midrange
- Quartiles ($Q_1, Q_2, Q_3$)
- Deciles ($D_1$ to $D_9$)
- Percentiles ($P_1$ to $P_{99}$)

# Statistical Measures and Tools (Part 2)

**Categories of Statistical Tools Used: Statistical Formulas**

- Central Moments ($\mu_r$)
- Raw Moments ($\mu'_r$)
- Skewness (Moments)
- Kurtosis (Percentiles)
- Excess Kurtosis (Moments)
- Covariance
- Pearson Skewness
- Bowley Skewness
- Correlation
- Regression byx ($b_{yx}$)
- Regression bxy ($b_{xy}$)
- Chebyshev's Inequality
- Normal PDF

# Statistical Measures and Tools (Part 3)

**Categories of Statistical Tools Used:**

**Statistical Concepts and Tools**

- PMF
- PDF
- CDF
- Expected Value
- Variance
- R-squared

**Inferential Statistics**

- Regression Analysis

**Visualization tools**

- Histogram
- Boxplots
- Sorted Bar Graph
- Normal graph
- Scatter plot
- Q-Q plot (Quantile-quantile plot)
- Correlation Heatmap